Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

October 11, 2026

1 Introduction

Who bears a tax, and how much of it passes through into the price, is the question of tax incidence (Weyl and Fabinger, 2013). A platform's commission is one example. Amazon deducts a referral fee from each sale, a commission of 15 percent of the sale price in many categories, so the seller receives the price net of the fee. This commission works like a sales tax, and we call it a tax from here on. Without returns, the pass-through and incidence of such a tax are well studied in the literature. Online sales, however, are often returned, with retailers expecting about one in five to come back.[1] On a kept sale the seller pays the whole tax. On a return the seller refunds the price in full and gets most of the tax back, but Amazon keeps part of it, at most 20 percent, or 3 percent of the price at a 15 percent rate.[2] Therefore the tax costs the seller different amounts on a kept sale and on a returned one. How much of this cost the seller can pass on to consumers depends on how purchases and returns respond to the price, and on how costly it is for the seller to handle a return. It is not clear how the pass-through and the burden borne by sellers and consumers change, or whether the platform should refund only part of the tax on a return. We study this in a competitive market, taking the tax schedule as given. Each buyer decides whether to buy and, after learning the fit, whether to keep the good, and the tax reaches both decisions only through the price.

A tax creates a wedge between what the buyer will pay and what the good costs the seller. Without returns, the wedge leads to underprovision on the purchase margin, because some beneficial trades do not happen. Returns add a second margin, the decision to keep the good, and a higher price can raise returns, because buyers who remain need a better fit to keep the good, or lower them, because the price changes who buys. It is therefore not clear whether the tax is still costly to society, as it is without returns, or in which direction the no-return results are biased when returns are ignored.

The paper asks who bears a tax that is partly refunded on returns and whether it is socially costly, for a small change in the tax. It takes four steps. First, how does the tax pass through into the price, and how do buyers and sellers divide the change in their own surplus? A rise in production cost and a rise in handling cost serve as benchmarks, since the first lowers the seller's margin on every unit, the second only on returned units, and the tax lies between the two. Second,

[1] National Retail Federation and Happy Returns, *2025 Retail Returns Landscape*, October 15, 2025, `https://nrf.com/research/2025-retail-returns-landscape`. The report estimates that 19.3 percent of online sales will be returned in 2025.

[2] Amazon, *Selling on Amazon pricing*, `https://sell.amazon.com/pricing`, and *North America selling fees*, `https://sell.amazon.com.sg/north-america/pricing`. Amazon calls the amount it keeps the refund administration fee, the lesser of five dollars or 20 percent of the referral fee.

counting the platform's revenue as part of social surplus, does any loss to society come from the decision to buy or from the decision to keep the good, and in which direction does that decision go wrong? Third, what is missed by ignoring returns and using the no-return results for this tax? Fourth, if the platform keeps a larger share of the tax on a refund, does society gain or lose? The answers depend only on the return rate and on how demand, supply, and returns respond to the price.

2 Setting

We study a homogeneous-good market under perfect competition with product returns, intermediated by a platform that charges sellers a referral fee. The product sells at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility of sending the good back, and in the distribution G_i of post-purchase match quality u. After buying, type i draws $u \sim G_i$. The consumer keeps the item when $u - p \geq -\kappa_i$, and then earns $u - p$. Otherwise the consumer returns the item and earns $-\kappa_i$. The refund equals the purchase price.

Expected surplus from buying at price p is

$$cs_i(p) = \mathbb{E}_{u \sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

Consumer i's purchase decision can be summarized by willingness to pay v_i, defined as the price solving $cs_i(v_i) = 0$. Type i buys if and only if $p \leq v_i$. Across types, v_i has distribution F with density f, so gross demand is $D(p) = 1 - F(p)$.

Consumer i's return probability is

$$\phi_i(p) = \Pr(u < p - \kappa_i).$$

Buyers with the same willingness to pay v can differ in κ_i and G_i, so they need not share a return probability. Their average return probability is

$$\phi(v, p) \equiv \mathbb{E}\big[\phi_i(p) \;\big|\; v_i = v\big],$$

which depends on v when the joint distribution of (κ_i, G_i) varies with willingness to pay. Averaging over buyers who purchase at p gives the aggregate return rate

$$r(p) = \frac{1}{D(p)} \int_p^\infty \phi(v, p)\, dF(v).$$

How the return rate changes with price splits into a behavioral term and a selection term,

$$r'(p) = \underbrace{\frac{1}{D(p)} \int_p^\infty \frac{\partial \phi(v,p)}{\partial p}\, dF(v)}_{\text{Behavioral channel}} + \underbrace{\frac{f(p)}{D(p)}\big(r(p) - \phi(p,p)\big)}_{\text{Selection channel}},$$

where $\phi(p, p)$ is the average return probability among types at the purchase margin $v_i = p$. The behavioral term is the change in return probability among buyers who continue to purchase after a price increase. A higher price raises their keep threshold, so they return more often. The selection term is the change in who buys. When $u = v_i + \varepsilon$, with ε and return hassle independent of v_i, selection is non-positive. Higher willingness to pay lowers the return probability, so buyers at the margin return more often than the average buyer and $\phi(p, p) \geq r(p)$. The two channels then move $r'(p)$ in opposite directions. Below, $r'(p)$ is left unsigned.

Under perfect competition, producer j takes the list price p and the return rate r as given and chooses output at cost $C_j(q_j)$, with strictly increasing marginal cost. The buyer pays the seller p, and the platform charges an ad valorem fee at rate τ. On a kept unit the seller keeps $(1-\tau)p$. On a returned unit the seller refunds the full price, so the buyer pays nothing. The seller pays net handling h on the good, which is gross handling cost minus scrap value and may be negative, and a share $\beta \in [0,1]$ of the referral fee is retained by the platform even when the product is returned. The seller therefore loses $\beta\tau p + h$ on that returned unit.[3] We maintain $p + h > 0$, which holds whenever scrap value does not exceed list price. With return rate r, expected payoff per unit sold is

$$(1-r)(1-\tau)p - r(\beta\tau p + h). \tag{1}$$

3 Consumer side

Consumer surplus at list price p integrates expected surplus over buyers. Write $cs(v,p)$ for expected surplus among types with willingness to pay v, so $cs(v,v) = 0$. Then

$$CS(p) = \int_p^\infty cs(v,p)\, dF(v). \tag{2}$$

The price derivative of consumer surplus equals minus kept demand. Differentiating in p gives an extensive-margin term, from types at the purchase margin, and an intensive-margin term, from surplus among inframarginal buyers,

$$CS'(p) = -cs(p,p)\, f(p) + \int_p^\infty cs_p(v,p)\, dF(v),$$

where $cs_p(v,p) \equiv \partial cs(v,p)/\partial p$. The extensive margin is zero because $cs(p,p) = 0$. On the intensive margin, a higher price lowers a buyer's surplus only when she keeps the good. Among buyers with willingness to pay v,

$$cs_p(v,p) = -\Pr(\text{keep} \mid v_i = v).$$

Aggregating over buyers gives

$$CS'(p) = \int_p^\infty cs_p(v,p)\, dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$.

A price path from p_0 to p_1 therefore changes consumer surplus by

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp.$$

A kept unit loses the price increase. A returned unit loses nothing. Buyers differ in how often they keep, so the same price increase cuts surplus by different amounts for different buyers. Those differences average into $r(p)$. $CS'(p)$ is then determined by $D(p)$ and $r(p)$, and ΔCS by the paths of those two aggregates.

Integrating that derivative from p up to a price $\bar{p}$ at which nobody buys gives the level of consumer surplus,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

[3] Amazon's refund administration fee is this rule for a referral fee. On a return Amazon keeps the lesser of \$5 or 20 percent of the referral fee, and gives the rest back to the seller. When the \$5 cap does not bind, $\beta = 0.2$.

4 Supply side

Set the fee aside. A gross sale brings revenue p if the consumer keeps the item, and it costs the refund p plus net handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit, the *effective margin*, is

$$\mu(p, h) \equiv p - r(p)(p + h). \tag{4}$$

Firm j takes p as given, so this margin enters the supply decision as a fixed number μ.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \ \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \ \ (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, equal to zero when μ is below its minimum marginal cost. Strictly increasing marginal cost implies $q_j'(\mu) > 0$ for active firms.

Market supply and industry profit sum these choices.

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Strictly increasing marginal cost implies $S'(\mu) > 0$, so quantity supplied rises with the effective margin. Market supply therefore falls in the list price whenever $\mu(p, h)$ falls in p.

Industry profit is the area to the left of market supply. The envelope theorem implies that firm j's profit rises with μ at rate $q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, that derivative integrates to

$$PS(\mu) = \int_0^{\mu} S(z)\, dz. \tag{6}$$

At equilibrium, $PS = \int_0^{\mu(p,h)} S(z)\, dz$.

5 A seller-remitted ad valorem tax

The expected payoff (1) is the seller's effective margin once the fee is included,

$$\mu^s(p, \tau, \beta, h) = (1 - r(p))(1 - \tau)p - r(p)(\beta\tau p + h). \tag{7}$$

The buyer's return decision depends only on p, so $r = r(p)$. The refund on a return is the list price, and the fee is remitted by the seller. Rewriting (7) in the pre-fee margin gives

$$\mu^s = \mu(p, h) - \tau p\big[1 - (1 - \beta)r(p)\big]. \tag{8}$$

The term in brackets is the share of the fee the seller expects to pay. That share is the whole fee on a kept unit, and the share β on a returned unit.

The fee reaches consumers only by moving the equilibrium list price. Market clearing is $D(p) = S(\mu^s)$. Consumer surplus depends on the list price. Producer surplus is the area under supply at μ^s,

$$CS = CS(p), \qquad PS = \int_0^{\mu^s} S(z)\, dz.$$

Tax derivatives are taken at $\tau = 0$. At that rate, the fee term in (8) has derivative zero with respect to p, so μ_p is the pre-fee derivative of $\mu(p, h)$. Holding p fixed, one percentage point lowers the margin by the expected fee share times the list price,

$$\mu_\tau^s = -p\big[1 - (1 - \beta)r(p)\big].$$

Write $Q \equiv D(p)$ and $\rho_\tau \equiv dp/d\tau$. Then

$$\frac{dCS}{d\tau} = -Q\big(1 - r\big)\rho_\tau, \qquad \frac{dPS}{d\tau} = Q\Big(\mu_p\rho_\tau - p\big[1 - (1 - \beta)r\big]\Big). \tag{9}$$

6 Pass-through

At a zero rate, $\mu^s = \mu(p, h)$, so equilibrium satisfies $D(p) = S(\mu(p, h))$. Write $Q \equiv D(p)$. A uniform one-dollar parallel shift in every firm's marginal production cost, indexed by c, shifts market supply at a given effective margin. A one-dollar increase in handling cost h lowers $\mu(p, h)$ at a fixed list price. The third shock is the ad valorem fee at rate τ.

For production and handling, market clearing is

$$m(p, x) \equiv D(p) - S(\mu(p, x), x) = 0, \qquad x \in \{c, h\}. \tag{10}$$

Differentiating $m = 0$ defines pass-through $\rho_x \equiv dp/dx$.

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

Expanding $m = D - S$ gives

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

hence

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{13}$$

The two shocks differ in which margin moves at fixed p or fixed μ.

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

The production term is $S_c = -S'(\mu)$. A one-dollar parallel rise in every firm's marginal cost is equivalent, at fixed μ, to evaluating market supply at $\mu - 1$. Handling lowers the effective margin at a fixed list price. Only returned units incur h, so a one-dollar rise lowers that margin by $r(p)$. Quantity supplied then falls by $r(p)\,S'(\mu)$, the fraction $r(p)$ of the production-cost numerator.

At a zero rate, pass-through of the fee equals $p\big[1-(1-\beta)r\big]$ times production-cost pass-through. The fee leaves the buyer's payment and the refund at p, so demand remains $D(p)$ and the return rate remains $r(p)$. At a fixed list price, the fee lowers the effective margin by $p\big[1 - (1 - \beta)r\big]$ dollars along an unchanged $S(\mu)$. Quantity supplied then falls by that amount times $S'(\mu)$. The same gap is a production-cost rise of $p\big[1 - (1 - \beta)r\big]$ dollars, so $\rho_\tau = p\big[1 - (1 - \beta)r\big]\rho_c$ at $\tau = 0$. With no returns, $r = 0$, so β drops out, the factor equals p, and pass-through is p times unit-cost pass-through (Weyl and Fabinger, 2013).

Result 1 (Pass-through into list price)**.** *Production-cost pass-through is*

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}. \tag{15}$$

The remaining shocks connect to this object by

$$\rho_h = r(p)\,\rho_c, \qquad\qquad \rho_\tau = p\big[1 - (1 - \beta)r(p)\big]\,\rho_c \quad \textit{at } \tau = 0. \tag{16}$$

6.1 Elasticity representation

The same pass-through formulas can be written in list-price elasticities. Write $\mu_p \equiv d\mu/dp$ for the response of the effective margin to the list price. Demand and supply elasticities with respect to list price are

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp}\frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p\,\frac{p}{S(\mu)}.$$

Here $\varepsilon_S(p)$ is the observed elasticity of market supply with respect to the clearing price. It has the sign of μ_p. With no returns, $\mu_p = 1$ and $\varepsilon_S > 0$. With returns, μ_p can be negative. Then $\varepsilon_S < 0$, and a higher list price reduces the quantity firms supply.

With no returns, pass-through equals $\varepsilon_S/(-\varepsilon_D + \varepsilon_S)$. Supply is then a function of the list price, so $S' = dS/dp$. A one-dollar cost increase cuts supply by S' units at the original price and opens an excess-demand gap of that size. A one-dollar rise in the list price closes the gap from both sides. Demand falls by $|D'|$ and supply rises by S'. Closing a gap of size S' therefore takes a price rise of $S'/(S' + |D'|)$.

Under the return margin, supply depends on the effective margin, so $S'(\mu) = dS/d\mu$. A one-dollar production-cost increase leaves μ unchanged at a fixed list price. It is equivalent to evaluating supply at $\mu - 1$, so supply falls by $S'(\mu)$ units and opens an excess-demand gap of that size. A one-dollar rise in the list price closes the gap at rate $S'(\mu)\,\mu_p - D'(p)$, so

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}.$$

A larger μ_p means that one dollar of list price restores more of the effective margin, and therefore more quantity supplied, so a smaller list-price movement closes the gap. Writing the same formula with ε_S places $1/\mu_p$ in the numerator,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{17}$$

The sign of ρ_c is the sign of the denominator $-\varepsilon_D + \varepsilon_S$. The numerator ε_S/μ_p is positive, because ε_S and μ_p have the same sign.

Result 2 (Negative pass-through)**.** *We have $\rho_c < 0$ if and only if $\varepsilon_S < 0$ ($\mu_p < 0$) and $|\varepsilon_S| > -\varepsilon_D$.*

A cost increase is then cleared by a lower list price. With $\varepsilon_S < 0$, so $\mu_p < 0$, a higher list price lowers the quantity firms supply. A rise in c shifts that supply curve to the left and creates excess demand at the initial p. A higher price would shrink supply further and widen the gap. The price falls. A lower p raises quantity demanded and, because supply slopes down, raises quantity supplied. When $|\varepsilon_S| > -\varepsilon_D$, supply is steeper than demand, so the supply response to the price cut exceeds the demand response, and the excess demand clears at a lower list price. In the classical model, $\mu_p = 1$, supply slopes up in the list price, and $\rho_c > 0$.

Proof. By (17),

$$\rho_c = \frac{\varepsilon_S/\mu_p}{-\varepsilon_D + \varepsilon_S}.$$

Since $\varepsilon_S = S'(\mu)\mu_p\, p/S(\mu)$, the terms ε_S and μ_p have the same sign, so $\varepsilon_S/\mu_p > 0$. Hence $\rho_c < 0$ if and only if

$$-\varepsilon_D + \varepsilon_S < 0,$$

which is equivalent to $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Because ε_S and μ_p have the same sign, $\varepsilon_S < 0$ is equivalent to $\mu_p < 0$. □

The effective margin rises by more than one dollar per dollar of list price when the return rate falls steeply enough. Differentiating $\mu = p - r(p)(p + h)$ gives

$$\mu_p = 1 - r(p) - r'(p)(p + h).$$

Write $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ for the elasticity of the return rate in list price. Then $\mu_p > 1$ if and only if $\varepsilon_r(p) < -p/(p + h)$. Selection must then outweigh the behavioral rise in individual return probabilities by enough that the aggregate return rate falls steeply in p. When the behavioral response dominates, ε_r is less negative, or positive, and $\mu_p < 1$.

The classical no-return formula treats the observed list-price slope of supply as if the effective margin moved one for one with price. That slope is $S'(\mu)\,\mu_p$, so the formula returns $\mu_p \rho_c$.

Result 3. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\mu_p > 1$, agrees with it when $\mu_p = 1$, and underestimates it when $0 \le \mu_p < 1$. When $\mu_p < 0$, the classical formula has the opposite sign of ρ_c. The condition $\mu_p > 1$ is equivalent to $\varepsilon_r(p) < -p/(p + h)$.*

When $\mu_p > 1$, one dollar of list price restores more than one dollar of effective margin, so a smaller price movement closes the excess-demand gap than the classical formula predicts. Higher handling cost widens the set of ε_r for which that overestimation occurs, because the threshold $-p/(p + h)$ moves toward zero. When $0 \le \mu_p < 1$, one dollar of list price restores less than one dollar of effective margin, so the list price must move by more than the classical formula predicts. When $\mu_p < 0$, the effective margin falls in the list price, and the classical formula predicts a price change of the wrong sign.

7 Local incidence

Production cost, handling cost, and the fee reach consumers only through the list price. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote equilibrium gross quantity and the return rate, and $\rho_x \equiv dp/dx$. Consumer surplus therefore changes by

$$\frac{dCS}{dx} = -Q(1 - r)\rho_x, \tag{18}$$

Producer surplus is $PS(\mu, x) = \int_0^\mu S(z, x)\, dz$. Differentiating in x and using the equilibrium condition $S(\mu, x) = Q$ gives

$$\frac{dPS}{dx} = Q\frac{d\mu}{dx} + \Sigma_x,$$

where $\Sigma_x \equiv \int_0^\mu (\partial S(z, x)/\partial x)\, dz$ is the direct shift in the area under the supply curve. The chain rule is $\frac{d\mu}{dx} = \mu_p \rho_x + \mu_x$, so

$$\frac{dPS}{dx} = Q(\mu_p \rho_x + \mu_x) + \Sigma_x. \tag{19}$$

The local incidence ratio is $I_x \equiv (dCS/dx)/(dPS/dx)$. For these three shocks,

$$I_x = \frac{-Q(1 - r)\rho_x}{Q(\mu_p \rho_x + \mu_x) + \Sigma_x},$$

and the rows of Table 1 reduce to one expression in ρ_c and μ_p.

Shock x	ρ_x	μ_x	Σ_x	dCS/dx	dPS/dx	I_x
Production cost c	ρ_c	0	$-Q$	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Handling cost h	$r\rho_c$	$-r$	0	$-Q(1-r)r\rho_c$	$Qr(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Fee τ	$\theta\rho_c$	$-\theta$	0	$-Q(1-r)\theta\rho_c$	$\theta Q(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$

Table 1: Pass-through, direct effects, surplus changes, and incidence ratio for each shock. The fee column is evaluated at a zero rate, with $\theta \equiv p\big[1-(1-\beta)r\big]$.

Result 4. *For production cost, handling cost, and the fee, the local incidence ratio is identical:*

$$I_x = \frac{(1-r)\rho_c}{1-\mu_p\rho_c} > 0. \tag{20}$$

Handling cost and the fee shift the effective margin by a scalar, and that scalar multiplies the changes in consumer and producer surplus. At a fixed list price a production-cost shock leaves μ unchanged. It shifts supply, and the producer loss before the price adjusts is Q, the same loss as a one-dollar fall in the effective margin. Handling shifts μ down by r dollars. At a zero rate the fee shifts it down by $p\big[1-(1-\beta)r\big]$ dollars. For a small shock that direct shift scales pass-through, so $\rho_h = r\rho_c$ and $\rho_\tau = p\big[1-(1-\beta)r\big]\rho_c$ at $\tau = 0$.

The scalar cancels in the incidence ratio. How far the equilibrium moves depends on the shock. How the movement splits between consumers and producers depends on the local slopes of demand and supply and on μ_p.[4]

The denominator $1-\mu_p\rho_c$ in (20) is the per-unit producer loss on a one-dollar cost shock. Before the price adjusts, producer surplus falls by Q. The list-price change moves the effective margin by $\mu_p\rho_c$ per unit, and the envelope $dPS/d\mu = Q$ turns that movement into a recovery of $Q\mu_p\rho_c$. The net change in producer surplus is $Q(\mu_p\rho_c - 1)$, so the loss per unit is $1-\mu_p\rho_c$. The numerator $(1-r)\rho_c$ is the per-unit consumer loss on kept units. The incidence ratio compares these two per-unit burdens. When the denominator is small and positive, consumers bear most of the loss. When the denominator is negative, the recovery exceeds the direct cost and producers gain.

Producer surplus rises with a production-cost increase exactly when pass-through is negative. The recovery factor equals $\mu_p\rho_c = \varepsilon_S/(\varepsilon_S - \varepsilon_D)$. It exceeds one if and only if $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Result 2 states that this is the condition for $\rho_c < 0$.

When pass-through is negative, clearing requires the effective margin to rise by more than one dollar. A one-dollar cost increase shifts the supply curve upward and opens an excess-demand gap at the initial price. Quantity supplied increases with the effective margin, so covering the original gap requires μ to rise by one dollar. The price fall also raises quantity demanded and opens a further gap. Supply must increase by more than the original shift, so the effective margin must rise by more than one dollar. Thus $\mu_p\rho_c > 1$ is the market-clearing requirement when pass-through is negative.

The incidence ratio is positive, so consumer and producer surplus move together. When $\rho_c < 0$, an increase in production cost, handling cost, or the fee lowers the list price. Consumer surplus rises, and producer surplus rises as well. When $\rho_c > 0$, all three shocks reduce both surpluses. In the classical model, $\mu_p = 1$ and pass-through is positive, so both surpluses fall.

[4]Consumers lose or gain through the list-price change ρ_x. Producers' receipt moves with the effective margin, which changes by $\mu_p\rho_x$ when the list price changes by ρ_x. Thus μ_p governs how much of each dollar of p feeds into producer surplus.

7.1 First-order private surplus and deadweight loss

Private surplus is the sum of consumer and producer surplus, $W \equiv CS + PS$. Adding (18) and (19) gives

$$\frac{dW}{dx} = -Q(1-r)\rho_x + Q(\mu_p\rho_x + \mu_x) + \Sigma_x = Q\big[\mu_x + \big(\mu_p - (1-r)\big)\rho_x\big] + \Sigma_x.$$

Substituting $\mu_p = 1 - r - r'(p)(p+h)$ gives

$$\frac{dW}{dx} = Q\big[\mu_x - r'(p)(p+h)\rho_x\big] + \Sigma_x, \tag{21}$$

with μ_x and Σ_x as in Table 1. The term $-Q\,r'(p)(p+h)\rho_x$ is the first-order effect of the price-induced change in the return rate. A change $r'(p)\rho_x$ in the return rate costs $p+h$ per unit sold. The term is zero when $r' = 0$.

With no returns, a unit production-cost increase lowers private surplus by Q. In that benchmark $r = 0$ and $r' = 0$, so (21) reduces to

$$\frac{dW}{dx} = Q\,\mu_x + \Sigma_x.$$

A unit production-cost increase shifts the supply curve up by one dollar at every quantity. The private-surplus loss is a rectangle of height one and width Q. Private surplus in that benchmark is $W = CS(p) + PS(p-c)$. Differentiating, the price terms $-Q\rho_c$ and $+Q\rho_c$ cancel, so $dW/dc = -Q$.

For an ad valorem tax at a zero rate and with no returns, $r = 0$, so β drops out. The same cancellation gives $dW/d\tau = -pQ$. One percentage point is a wedge of p dollars on every unit, and the price-change terms cancel. Platform revenue $G = \tau pQ$ is a transfer. At $\tau = 0$, $dV/d\tau = dW/d\tau + pQ = 0$. A one-dollar cost shock destroys Q dollars of private surplus. An infinitesimal fee moves pQ dollars from private surplus to the platform, so social surplus is unchanged at first order.

With returns, the return-rate term in (21) is generally nonzero for a production-cost increase, a handling-cost increase, and the the fee. Private surplus then changes through the keep-or-return margin as well as through the direct wedge. For the fee, platform revenue is still a transfer. It is added back below, with $G = \tau p\big[1 - (1-\beta)r\big]Q$.

Substituting μ_x and Σ_x from Table 1 into (21), and using the pass-through links in Result 1, gives the following derivatives. The tax derivative is evaluated at a zero rate.

$$\frac{dW}{dc} = -Q\,\Lambda, \qquad \frac{dW}{dh} = -Q\,r\,\Lambda, \qquad \frac{dW}{d\tau} = -p\big[1-(1-\beta)r\big]Q\,\Lambda, \tag{22}$$

where

$$\Lambda \equiv 1 + r'(p)(p+h)\,\rho_c. \tag{23}$$

The unit term in Λ is the direct wedge on units already sold. For a production-cost shock that wedge is one dollar. For the fee it is $p\big[1-(1-\beta)r\big]$ dollars. List-price terms cancel, so the change in the number of buyers does not enter at first order. The term $r'(p)(p+h)\rho_c$ is the further change on the keep-or-return margin after the price moves. Each change $r'(p)\rho_c$ in the return rate costs $p+h$ per unit sold. A handling-cost shock multiplies the production-cost derivative by r.

Freeze the aggregate return rate at its initial value $r_0 \equiv r(p_0)$. In that ex-ante economy the effective margin is (4) with $r(p)$ replaced by the constant r_0,

$$\mu^{\mathrm{EA}}(p) \equiv p - r_0(p+h). \tag{24}$$

Applying (21) to the production-cost shock in this economy yields $dW/dc\big|_{\text{EA}} = -Q$, the direct cost on units sold with the return rate held at r_0. The ex-post piece is the residual $dW/dc\big|_{\text{EP}} \equiv dW/dc - dW/dc\big|_{\text{EA}}$. Since $dW/dc = -Q\Lambda$,

$$\frac{dW}{dc}\Big|_{\text{EP}} = -Q\, r'(p)(p+h)\rho_c. \tag{25}$$

For the fee at a zero rate, both pieces are multiplied by $p\big[1-(1-\beta)r\big]$. The ex-ante piece is $-p\big[1-(1-\beta)r\big]Q$. The ex-post piece is $-p\big[1-(1-\beta)r\big]Q\, r'(p)(p+h)\rho_c$.

Result 5 (Ex-ante and ex-post decomposition)**.** *For a unit production-cost shock,*

$$\frac{dW}{dc} = -Q - Q\, r'(p)(p+h)\rho_c. \tag{26}$$

The first term is the ex-ante change on the purchase margin. The second is the ex-post change on the keep-or-return margin. Each unit switched from keep to return costs $p+h$. For the fee at a zero rate both terms are multiplied by $p\big[1-(1-\beta)r\big]$. For a handling shock the two terms become $-Qr$ and $-Qr\, r'(p)(p+h)\rho_c$. The ex-ante term is the direct cost on units already sold. The change in the number of buyers does not enter at first order, so this term is not a loss of purchases. The ex-post term is a further loss when $r'(p) > 0$ and $\rho_c > 0$: the higher price raises the return rate, and there are too many returns. It is a gain, with private surplus still falling, when $r'(p) < 0$: pass-through is then positive, the higher price lowers the return rate, and there are fewer returns than at the original price. It is a gain that exceeds the ex-ante loss when $\rho_c < 0$: the price fall cuts returns, and private surplus rises.

Consumer and producer surplus move together, so Λ has the same sign as ρ_c. The no-return formula keeps only the ex-ante rectangle $-Q$. The gap between that formula and $dW/dc = -Q\Lambda$ is the ex-post term $-Q\, r'(p)(p+h)\rho_c$. That term can offset the rectangle, amplify it, or reverse the sign of the private-surplus change.

When selection dominates, $r'(p) < 0$. Then $\mu_p > 0$ and hence $\rho_c > 0$, so a cost increase raises the list price. The higher price screens out frequent returners and lowers the return rate. The ex-post term is then a gain, and $0 < \Lambda < 1$. The private-surplus change stays negative, because $\rho_c > 0$ keeps consumer and producer surplus falling.

When $r'(p) > 0$ and $\rho_c > 0$, the higher list price raises the return rate. Extra returns cost $p+h$ each, so the ex-post term amplifies the rectangle and $\Lambda > 1$. That amplification is larger when $r'(p)$ or h is large.

When $\rho_c < 0$, a production-cost, handling-cost, or fee increase lowers the list price, so consumer surplus rises. The incidence ratio is positive, so producer surplus rises as well, a Pareto improvement. Negative pass-through requires $r'(p) > 0$, so the fall in the list price lowers the return rate. Each unit that switches from return to keep saves $p+h$, and that saving more than offsets the direct wedge. Private surplus rises by $Q|\Lambda|$ under the cost shock and by $p\big[1-(1-\beta)r\big]Q|\Lambda|$ under the fee, with $\Lambda < 0$. The gain is larger when $r'(p)(p+h)\,|\rho_c|$ is large. Result 6 records these three comparisons.

Result 6 (No-return private-surplus bias)**.** *Consider a unit production-cost increase. Relative to the correct returns-adjusted change $dW/dc = -Q\Lambda$, the no-return formula $dW/dc = -Q$*

(i) overestimates the private-surplus loss if and only if $r'(p) < 0$ (which implies $\mu_p > 0$ and hence $\rho_c > 0$, so $0 < \Lambda < 1$);

(ii) underestimates the private-surplus loss if and only if $r'(p) > 0$ and $\rho_c > 0$ (hence $\Lambda > 1$);

(iii) overestimates the private-surplus loss with the wrong sign—reporting a loss of Q when the shock in fact raises private surplus by $Q|\Lambda|$—if and only if $r'(p) > 0$ and $\rho_c < 0$.

For the fee at a zero rate the same three comparisons apply to the ex-ante benchmark $-p\big[1-(1-\beta)r\big]Q$, which holds the return rate fixed. A formula that also sets the return rate to zero uses $-pQ$. The two agree when $\beta = 1$ or $r = 0$. When $\beta < 1$ and $r > 0$, $-pQ$ is a larger private loss, because the ex-ante benchmark gives the seller back the share $1-\beta$ of the fee on each returned unit.

Social surplus adds back the revenue the platform keeps. On a kept unit that revenue is τp. On a returned unit it is the share $\beta\tau p$. Expected revenue per unit is therefore $\tau p\big[1-(1-\beta)r\big]$, and

$$G = \tau p\big[1-(1-\beta)r\big]Q,$$

with Q the equilibrium gross quantity. Write $V = W + G$. Differentiating at $\tau = 0$ gives $dG/d\tau = p\big[1-(1-\beta)r\big]Q$, so $dV/d\tau = dW/d\tau + p\big[1-(1-\beta)r\big]Q$. Using (22) gives the change in social surplus.

Result 7 (Social surplus)**.** *At a zero rate,*

$$\frac{dV}{d\tau} = -p\big[1-(1-\beta)r\big]Q(\Lambda-1) = -p\big[1-(1-\beta)r\big]Q\,r'(p)(p+h)\rho_c. \tag{27}$$

An infinitesimal fee raises social surplus when $r'(p)\rho_c < 0$ and lowers it when $r'(p)\rho_c > 0$. The factor $p\big[1-(1-\beta)r\big]$ scales this change, so a larger share of the fee kept on a refund enlarges the gain or the loss.

8 Global Incidence

The fee row of Table 1 is the derivative at a zero rate. It is not the integrand for a finite change in τ.

A finite production-cost or handling-cost change integrates that shock's row in the same table, with the row re-evaluated at the current equilibrium.

Along such a path, the global incidence ratio $I_{x_0}^{x_1} \equiv \Delta CS/\Delta PS$, when $\Delta PS \neq 0$, is the weighted average of the local ratio implied by $dCS/dx = I(x)\,dPS/dx$,

$$I_{x_0}^{x_1} = \int_{x_0}^{x_1} I(x)\,\omega(x)\,dx, \tag{28}$$

where $\omega(x) \equiv (dPS/dx)/\Delta PS$.

The global ratio can be negative, even though $I(x)$ is positive at every point on the path. If dPS/dx changes sign along the path, the weights $\omega(x)$ take both signs, and the weighted average of positive local ratios can be negative. A finite production-cost or handling-cost change therefore need not preserve the sign of local incidence.

A Welfare bias in primitives

Result 6 has an equivalent statement in elasticities. Recall $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ and $\mu_p = 1 - r(p) - r'(p)(p+h)$. Then $r'(p)(p+h) = \varepsilon_r(p)\, r(p)\, (p+h)/p$, so

$$\mu_p > 0 \iff \varepsilon_r(p) < \frac{p}{p+h}\,\frac{1-r(p)}{r(p)}, \tag{29}$$

In these elasticities, Result 6 reads as follows.

(i) overestimation iff $\varepsilon_r(p) < 0$;

(ii) underestimation iff $\varepsilon_r(p) > 0$ and either $\mu_p \geq 0$, or both $\mu_p < 0$ and $|\varepsilon_S| < -\varepsilon_D$;

(iii) overestimation with wrong sign (predicted loss, actual gain) iff $\mu_p < 0$ and $|\varepsilon_S| > -\varepsilon_D$.

Proof. By Result 6, overestimation with $0 < \Lambda < 1$ is $r' < 0$, underestimation is $r' > 0$ with $\rho_c > 0$, and overestimation with wrong sign is $r' > 0$ with $\rho_c < 0$. Since ε_r has the same sign as r', write these in ε_r. For (i), $\varepsilon_r < 0$ forces $\mu_p > 1 - r > 0$ and hence $\rho_c > 0$, so $0 < \Lambda < 1$. When $\mu_p = 0$ and $r(p) < 1$, the derivative $r'(p)$ is positive, $\varepsilon_S = 0$, and $\rho_c = S'(\mu)/(-D'(p)) > 0$, so that case is in (ii). Result 2 translates the sign of ρ_c into conditions on μ_p and $|\varepsilon_S| \gtrless -\varepsilon_D$: with $\varepsilon_r > 0$ this is (ii), and $\rho_c < 0$ is (iii). □

References

E. Glen Weyl and Michal Fabinger. Pass-through as an economic tool: Principles of incidence under imperfect competition. *Journal of Political Economy*, 121(3):528–583, 2013.